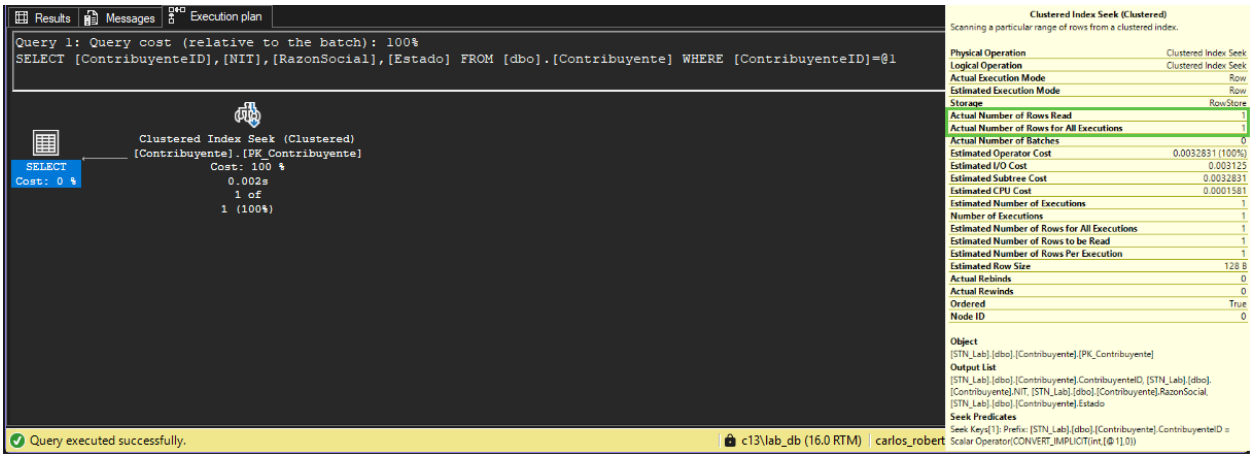
05_seek_vs_scan_impact

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos

5.1 CONTROLLED SEEK: ONE ROW

```sql
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente
WHERE ContribuyenteID = 500;

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



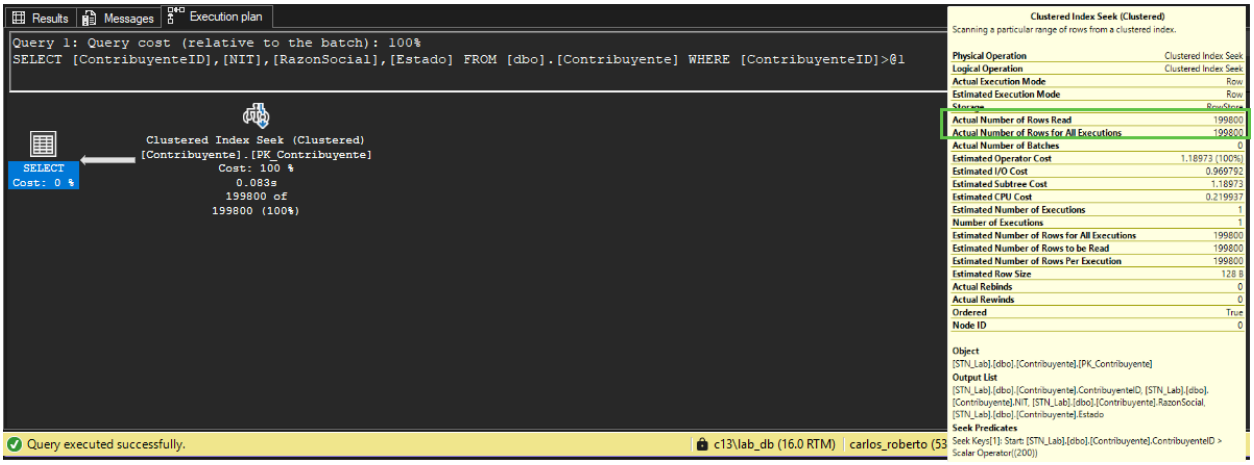
OBSERVACION 5.1

Reads_5.1 = *3*

(Clustered Index Seek 1 row)

5.2 CONTROLLED SCAN: ALL ROWS

```
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT ContribuyenteID, NIT, RazonSocial, Estado
FROM dbo.Contribuyente
WHERE ContribuyenteID > 0;

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



OBSERVACIÓN 5.2

- reads_5.2 = 1314 (Clustered Index Scan all rows)
- Ratio seek/scan = reads_5.2 / reads_5.1 = 438x

El optimizador eligio Seek sobre ContribuyenteID > 0 porque es la clave clustered - navega el B-tree desde el primer valor y lee en orden.

5.3 THE TIPPING POINT

```sql
-- 5.3.1 -- 0.5% of rows (1,000 rows)
SET STATISTICS IO ON;
SELECT ContribuyenteID, NIT, RazonSocial
FROM dbo.Contribuyente
WHERE ContribuyenteID BETWEEN 1 AND 1000;
SET STATISTICS IO OFF;
GO

-- 5.3.2 -- 5% of rows (10,000 rows)
SET STATISTICS IO ON;
SELECT ContribuyenteID, NIT, RazonSocial
FROM dbo.Contribuyente
WHERE ContribuyenteID BETWEEN 1 AND 10000;
SET STATISTICS IO OFF;
GO

-- 5.3.3 -- 15% of rows (30,000 rows)
SET STATISTICS IO ON;
SELECT ContribuyenteID, NIT, RazonSocial
FROM dbo.Contribuyente
WHERE ContribuyenteID BETWEEN 1 AND 30000;
SET STATISTICS IO OFF;
GO

-- 5.3.4 -- 30% of rows (60,000 rows)
SET STATISTICS IO ON;
SELECT ContribuyenteID, NIT, RazonSocial
FROM dbo.Contribuyente
WHERE ContribuyenteID BETWEEN 1 AND 60000;
SET STATISTICS IO OFF;
GO
```

OBSERVACION 5.3

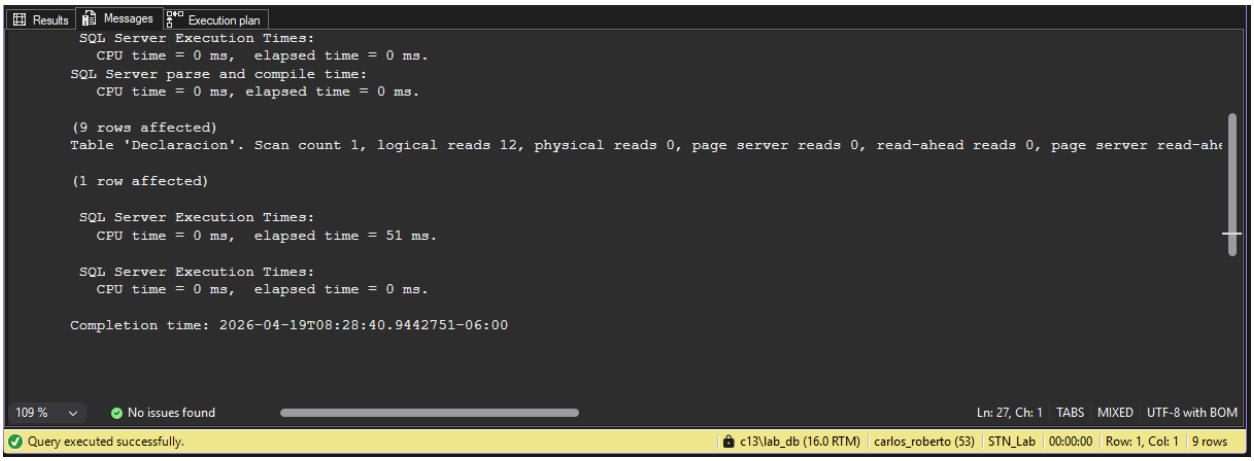
Range	Rows	Operator	Reads
1-1,000	1,000	Clustered Index Scan	536

1-10,000	10,000	Clustered Index Scan	536
1-30,000	30,000	Clustered Index Scan	536
1-60,000	60,000	Clustered Index Scan	536

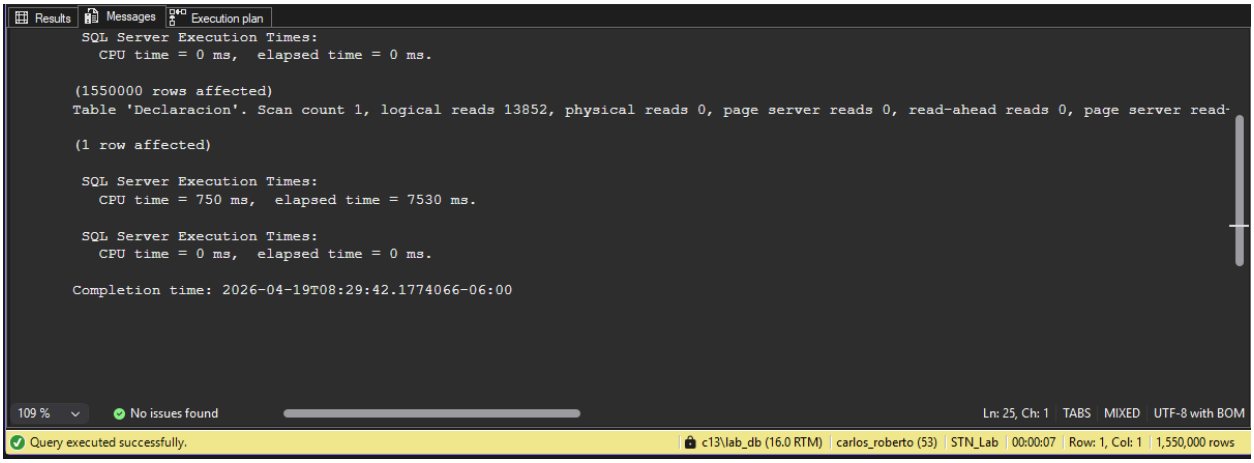
Conclusion: los reads son identicos en los 4 rangos.
El optimizador decidio que leer todas las paginas del clustered era mas barato que hacer Seek + Key Lookup por NIT y RazonSocial que estan fuera de la clave.
El tipping point ocurrio antes del 0.5% de las filas.
Con indices cubiertos el comportamiento cambiaria.

5.4 SEEK vs SCAN ON DECLARACION

```sql
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT DeclaracionID, ContribuyenteID, PeriodoFiscal, MontoDe
clarado
FROM dbo.Declaracion
WHERE ContribuyenteID = 500;

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



```sql
SET STATISTICS IO ON;
SET STATISTICS TIME ON;

SELECT DeclaracionID, ContribuyenteID, PeriodoFiscal, MontoDe
clarado
FROM dbo.Declaracion WITH (INDEX = 0);

SET STATISTICS IO OFF;
SET STATISTICS TIME OFF;
GO
```



reads_5.4.1 = 12

reads_5.4.2 = 13852

Ratio = reads_5.4.2 / reads_5.4.1 = 1154.33x

5.5 STATISTICS HEALTH CHECK

```sql
SELECT
    t.name                                  AS table_name,
    s.name                                  AS statistics_name,
    sp.last_updated,
    sp.rows,
    sp.rows_sampled,
```

```sql
    CAST(sp.rows_sampled * 100.0
         / NULLIF(sp.rows, 0) AS DECIMAL(5,2)) AS sample_pct,
    sp.modification_counter
FROM sys.tables t
JOIN sys.stats s
    ON t.object_id = s.object_id
CROSS APPLY sys.dm_db_stats_properties(s.object_id, s.stats_i
d) sp
WHERE t.name IN ('Contribuyente', 'Declaracion')
ORDER BY t.name, sp.last_updated DESC;
GO
```

Results | Messages | Execution plan

	table_name	statistics_name	last_updated	rows	rows_sampled	sample_pct	modification_counter
1	Contribuyente	_WA_Sys_00000007_35BCFE0A	2026-03-29 08:46:12.0666667	200000	171404	85.70	0
2	Contribuyente	_WA_Sys_00000004_35BCFE0A	2026-03-22 08:17:27.6766667	200000	169191	84.60	0
3	Contribuyente	_WA_Sys_00000003_35BCFE0A	2026-03-08 08:12:32.5100000	200000	171404	85.70	0
4	Contribuyente	_WA_Sys_00000005_35BCFE0A	2026-03-08 08:12:32.2633333	200000	169191	84.60	0
5	Contribuyente	IX_Contribuyente_NIT	2026-03-08 07:18:19.9700000	200000	200000	100.00	0
6	Contribuyente	PK_Contribuyente	2026-03-08 07:10:43.6233333	200000	200000	100.00	0
7	Contribuyente	_WA_Sys_00000002_35BCFE0A	2026-03-03 00:37:07.4233333	200000	172524	86.26	0
8	Declaracion	IX_Declaracion_Fecha_Estado	2026-04-12 12:57:50.7833333	1550000	1550000	100.00	0
9	Declaracion	IX_Declaracion_Estado_Fecha	2026-04-12 12:54:33.0600000	1550000	1550000	100.00	0
10	Declaracion	IX_Declaracion_Periodo_Contribuyente	2026-04-12 12:47:50.0500000	1550000	1550000	100.00	0
11	Declaracion	IX_Declaracion_Contribuyente_Periodo	2026-04-12 12:34:32.2333333	1550000	1550000	100.00	0
12	Declaracion	IX_Declaracion_ContribuyenteID	2026-04-12 12:30:25.3200000	1550000	1550000	100.00	0
13	Declaracion	_WA_Sys_00000008_36B12243	2026-03-29 05:11:27.1600000	1550000	147085	9.49	0
14	Declaracion	_WA_Sys_00000005_36B12243	2026-03-29 05:11:27.0500000	1550000	152618	9.85	0
15	Declaracion	_WA_Sys_00000004_36B12243	2026-03-22 09:10:15.5766667	1550000	1550000	100.00	0
16	Declaracion	_WA_Sys_00000002_36B12243	2026-03-22 09:10:14.7900000	1550000	1550000	100.00	0
17	Declaracion	_WA_Sys_00000003_36B12243	2026-03-22 09:10:14.4800000	1550000	1550000	100.00	0
18	Declaracion	PK_Declaracion	2026-03-22 09:10:14.1600000	1550000	1550000	100.00	0

Query executed successfully. | c13\lab_db (16.0 RTM) | carlos_roberto (53) | STN_Lab | 00:00:00 | Row: 1, Col: 1 | 18 rows

OBSERVACION 5.5

- Si modification_counter supera el 20% del total de filas las estadisticas estan desactualizadas.
- Ejecutar:

```sql
UPDATE STATISTICS dbo.Contribuyente WITH FULLSCAN;
```